

Manipulação com ∞

- $\pm \infty \pm \infty = \pm \infty$
- $(+\infty) - (+\infty) = \text{indeterminado}$
- $+\infty + k = +\infty$
- $-\infty + k = -\infty$
- $(+\infty) \cdot (+\infty) = +\infty$
- $(+\infty) \cdot (-\infty) = -\infty$
- $+\infty \cdot k = +\infty \quad k > 0$
- $+\infty \cdot k = -\infty \quad k < 0$
- $\pm \infty \cdot 0 = \text{indeterminado}$
- $k / \pm \infty = 0$
- $\pm \infty / \pm \infty = \text{indeterminado}$
- $k / 0^+ = +\infty, k > 0$
- $+\infty / 0^+ = +\infty$
- $k / 0^- = -\infty, k > 0$
- $+\infty / 0^- = -\infty$
- $0 / 0 = \text{indeterminado}$

Exemplo:

Limites Infinitos de Funções Racionais

$$a) \lim_{x \rightarrow +\infty} \frac{2x-5}{x+8} =$$

$$\lim_{x \rightarrow +\infty} \frac{2x-5 : x}{x+8 : x} = \frac{\frac{2x}{x} - \frac{5}{x}}{\frac{x}{x} + \frac{8}{x}} \rightarrow \lim_{x \rightarrow \infty} \frac{2 - \frac{5}{x} \rightarrow 0}{1 + \frac{8}{x} \rightarrow 0}$$

$$= \frac{2}{1} = 2.$$

$$b) \lim_{x \rightarrow +\infty} \frac{2x+5}{\sqrt{2x^2-5}} = \lim_{x \rightarrow +\infty} \frac{\frac{2x}{x} + \frac{5}{x} \rightarrow 0}{\sqrt{\frac{2x^2}{x^2} - \frac{5}{x^2} \rightarrow 0}} \rightarrow \frac{2}{\sqrt{2}} = \sqrt{2}.$$